# Markov Chain Lock Watermarking: A Calibrated Policy-to-Configuration Map for EU AI Act Article 50

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## Abstract

Regulatory regimes such as the EU AI Act mandate machine-readable marking of synthetic text, but existing watermark detectors rely on access to the generating language model and on entropy-based heuristics with no closed-form calibration to a target false-positive rate. We introduce *Markov Chain Lock* (MCL), an active watermark that partitions the vocabulary into $S$ states via keyed SHA-256 hashing and forces a hard Markov transition on a fraction $\rho$ of generated positions; the detector replays the partition from the same secret key and is therefore model-free, running in $O(n)$ hash operations. Theoretically, we derive an operator-facing formula mapping a target false-positive rate, text length, and budget to the minimum state count $S^\star$ (Theorem A.2). We further prove a universal robustness threshold $\delta^\star = 1 - 1/\sqrt{2} \approx 29.3\%$ that is independent of $(S, \rho, n)$ (Theorem A.3). Both results adopt the same midpoint decision convention used throughout the paper, and generalise to any $k$-regular transition topology (Theorem A.5). Empirically, we compare MCL head-to-head against KGW and SWEET across multiple instruction-tuned LLMs and four text domains under matched-budget translation and random-substitution attacks. We then sweep the state count $S$ to anchor the calibration formula on data and validate the $k$-regular generalisation at $k = 2$ (section 6). The closed-form calibration and model-free detector together let a third-party auditor verify a marked deployment from the text and the (publicly known) tokenizer alone, without LM access.


## 1. Introduction

EU AI Act Article 50 (European Parliament and Council, 2024) requires a machine-readable mark on AI-generated text by August 2026, but does not specify which false-positive rate, text length, or quality cost constitute a compliant detector. Operators therefore need a watermark whose detector is auditable end-to-end. A regulator or third-party verifier should be able to confirm provenance from the text, the secret key, and the (publicly known) tokenizer alone — without query access to the generating language model. The widely used logit-bias schemes do not meet this bar. KGW (Kirchenbauer et al., 2023) and its entropy-gated variants such as SWEET (Lee et al., 2024) and EWD (Lu et al., 2024) require the language model (or a faithful proxy) at detection time in order to recompute the green-list bias or to weight tokens by entropy. Undetectability-frontier results (Christ et al., 2024) characterise what any scheme can achieve information-theoretically but yield no deployable construction, and passive detectors (Mitchell et al., 2023; Tian, 2023) expose no controllable false-positive frontier at all.

**Markov Chain Lock.** We propose *Markov Chain Lock* (MCL), an active watermark whose detector is model-free. A keyed SHA-256 hash partitions the vocabulary into $S$ equivalence classes (states), and a fixed cycle over those states defines a Markov transition pattern. On a fraction $\rho$ of generated positions, the language-model logits are masked to keep only tokens whose state is the legal successor of the current state; the produced text therefore walks the chain by construction. Detection re-derives every token's state from the key in $O(n)$ hash operations and counts how often consecutive tokens follow the cycle. Because the state map is a function of the token id and the key only, no language-model forward pass, weights, or proxy LM is required to verify a transcript. Two operational consequences follow. First, third-party audit becomes lightweight: a regulator with the key can verify provenance from a string of plain text. Second, the detector's null-and-alternative law inverts in closed form to a calibration map (Theorem A.2, derived in section A). A regulator who fixes a target false-positive rate $\alpha$, a text-length floor $n_{\min}$, and a watermark budget $\rho$ reads off the minimum admissible state count $S^\star(n_{\min}, \rho, \alpha)$; a

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deployer may spend more budget than the floor for additional margin but cannot evade the regulator's specification. Figure 1 summarises the construction; the formal treatment is in section 4 and section 5.

**Gating as a design dial.** A second design choice — *which* of the $n$ positions to watermark — is left as a parameter we call the gate. Existing schemes implicitly fix this choice: SWEET and EWD watermark only high-entropy positions, on the intuition that those positions contribute the bulk of the detection statistic. We adopt the high-entropy gate by default for parity with SWEET and compare MCL against KGW and SWEET at matched budget on multiple instruction-tuned LLMs across four text domains under three attack conditions (*clean*, random substitution, and ZH back-translation; section 6).

**Contributions.**

1. **A model-free watermarking primitive.** We introduce MCL, a discrete hard-constraint watermark with cryptographic state partitions, clockwork transitions, and $O(n)$ model-free detection (section 4).
2. **Closed-form calibration.** We derive an operator-facing calibration map taking a target false-positive rate, text-length floor, and budget to the minimum admissible state count (Theorem A.2, section A), and a universal robustness threshold that is invariant to $(S, \rho, n)$ (Theorem A.3). Both adopt the midpoint decision convention used throughout the paper.
3. **$k$-regular generalisation.** Both the detection law and the robustness threshold extend verbatim to any $k$-regular doubly-stochastic transition topology (Theorem A.5); clockwork is the $k{=}1$ instance.
4. **Empirical head-to-head comparison.** We evaluate MCL against KGW and SWEET at matched budget across multiple instruction-tuned language models and four text domains under translation and random-substitution attacks (section 6), and show that MCL retains substantially more detection signal than the baselines (subsection 6.2, subsection D.3).
5. **Empirical anchor of the calibration formula.** An independent state-count sweep recovers the closed-form $S^\star$ ordering to within one state and motivates the empirical-SD threshold recalibration that closes the residual FPR gap (subsection 6.5, subsection D.2).

**Roadmap.** section 2 situates MCL within active and passive watermarking. section 3 fixes notation; section 4 specifies the construction; section 5 states the calibration, robustness, and $k$-regular results; and section 6 reports the empirical comparison. Proofs and additional derivations are in section A.

## 2. Related Work

**Passive detection.** DetectGPT (Mitchell et al., 2023) exploits log-probability curvature; GPTZero (Tian, 2023) combines perplexity and burstiness. Both read statistical traces in existing text, with no controllable false-positive frontier and sharp degradation under paraphrase (Krishna et al., 2023; Sadasivan et al., 2023). They cannot serve as the technical substrate for Article 50 because neither regulator nor deployer can set the detection regime in advance.

**Active watermarking.** Green–red-list watermarking (Kirchenbauer et al., 2023) biases next-token logits toward a hashed green list and detects via a $z$-test on green-token frequency. The scheme requires temperature sampling and degrades under paraphrase, and offers no analytic inversion from a regulatory parameter back to a bias magnitude. MCL's fingerprint $\phi$ is structurally a Bernoulli$(1/S)$ $z$-test, so the calibration formula (8) is the analogue KGW lacks.

**Distortion-free schemes.** Aaronson (2023)'s Gumbel-max construction and Kuditipudi et al. (2024)'s inverse-transform sampling watermark are *distortion-free*: they preserve the LM marginal in expectation over the key. Christ et al. (2024) prove cryptographic indistinguishability under sufficient entropy. These schemes beat MCL on quality strictly (zero expected KL versus MCL's $\geq \rho \log S$). The trade is that they require either LM access or the key sequence at detection time; MCL is the distorting alternative that buys model-free third-party audit and the closed-form $(\alpha, n, \rho) \mapsto S^\star$ map.

**Entropy-adaptive schemes: gates vs. detectors.** SWEET (Lee et al., 2024) and EWD (Lu et al., 2024) both restrict watermarking signal to *high-entropy* positions, but at different layers. SWEET applies a binary gate at generation time (restricting where the green-list bias is added); EWD weights positions by entropy at detection time, approximating the log-likelihood ratio of Theorem A.8. MCL is gate-agnostic in its mathematical specification: high-entropy, low-entropy, and surprisal-gap gates are all admissible. For the empirical comparison in section 6 we adopt the high-entropy gate, matching SWEET's gating policy at the same budget; the operational dial we vary in our matched-budget head-to-head is therefore $\rho$, not the gate identity.

**Technical AI governance.** Article 50 of the EU AI Act (European Parliament and Council, 2024) and the EU's draft Code of Practice on Transparency (European Commission AI Office, 2025) require machine-readable marking of AI-generated content; the OECD Hiroshima Process (OECD, 2024) provides a parallel international frame-

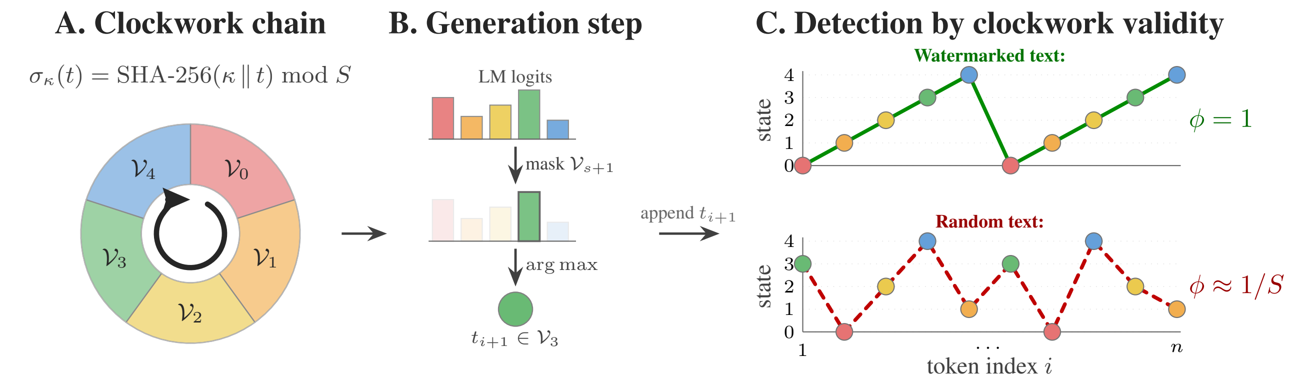



*Figure 1.* **Markov Chain Lock at a glance. (A)** A keyed SHA-256 hash $\sigma_\kappa$ partitions the vocabulary into $S$ disjoint sets $\mathcal{V}_0, \dots, \mathcal{V}_{S-1}$ ordered into a clockwork cycle $s \to s{+}1 \bmod S$ (here $S{=}5$). **(B)** At a watermarked position with current state $s$, the language-model logits are masked to keep only the next partition $\mathcal{V}_{s+1}$, and the next token is the masked $\arg\max$. **(C)** Detection re-derives every token's state from the key $\kappa$ in $O(n)$ hash operations: a watermarked sequence walks the chain by construction (top, validity rate $\phi = 1$), whereas unwatermarked text visits states uniformly and is valid only at rate $1/S$ in expectation (bottom).

work. These instruments leave the detection regime (FPR, quality floor, robustness) implicit. MCL's contribution, beyond the watermarking scheme itself, is that it surfaces these parameters analytically, producing a policy-to-configuration map auditable by third parties (section 7).

## 3. Preliminaries

**Notation.** Let $\mathcal{V}$ denote the vocabulary of an autoregressive language model, $V = |\mathcal{V}|$, and write $\mathbf{t} = (t_1, \dots, t_n) \in \mathcal{V}^n$ for a token sequence of length $n$. The deployer and the verifier share a secret key $\kappa \in \{0,1\}^*$ and the same tokenizer; the verifier never sees the language model. Let $\mathcal{H} : \{0,1\}^* \to \{0,1\}^{256}$ be SHA-256, modelled as a random oracle in the security analysis (Theorem A.7). We write $[m] = \{0, \dots, m-1\}$ and $z_\alpha$ for the one-sided standard-normal quantile at level $\alpha$.

**Threat model.** The adversary observes watermarked text and may paraphrase, translate, edit, splice, or delete tokens before publication; we model these as a per-token modification rate $\delta \in [0, 1)$. The adversary is computationally bounded and has no access to $\kappa$ or to the language model at detection time. The verifier sees only token text and runs in $O(n)$ hash operations.

**Detection problem and quality.** Detection is a binary hypothesis test on text alone—$H_0$ (non-watermarked) vs. $H_1$ (watermarked under key $\kappa$)—at a fixed false-positive rate $\alpha$, since false positives are operationally the costly error. The watermark distribution must remain fluent under standard greedy or low-temperature decoding, which we control via the per-token KL divergence to the original next-token distribution and report as perplexity inflation (section 4).

## 4. The MCL Scheme

### 4.1. State partition

For each token id $v \in \mathcal{V}$ define

$$\sigma_\kappa(v) \;=\; \mathcal{H}(\kappa \,\|\, v) \bmod S, \tag{1}$$

which deterministically maps $\mathcal{V}$ onto $S$ equivalence classes $\mathcal{V}_s = \{v : \sigma_\kappa(v) = s\}$, $s \in [S]$. Under the random-oracle model $\sigma_\kappa$ is uniform and independent across distinct tokens, so $\mathbb{E}[|\mathcal{V}_s|] = V/S$; knowledge of $\sigma_\kappa$ on queried tokens reveals nothing about unqueried ones (Theorem A.7).

### 4.2. Transition topology

A successor function $\Sigma_k : [S] \to \binom{[S]}{k}$ assigns each state a $k$-element set of legal next states. We require $\Sigma_k$ to be $k$-regular in both out- and in-degree; column-regularity is essential for the null-variance and robustness arguments (Theorem A.5, Theorem A.1). The default is the *clockwork* chain at $k = 1$, $\Sigma_1(s) = \{(s{+}1) \bmod S\}$: deterministic, periodic of period $S$, uniform stationary distribution, and the smallest random baseline $1/S$ among 1-regular topologies. We additionally evaluate the *soft-cycle* variant $k = 2$, $\Sigma_2(s) = \{(s{+}1) \bmod S, (s{+}2) \bmod S\}$, which doubles the per-state successor budget at the cost of raising the random baseline to $2/S$ (section 6).

### 4.3. Generation

At step $i$ with current state $s_i = \sigma_\kappa(t_i)$, MCL forms the legal token set $\mathcal{V}^\star_{s_i} = \bigcup_{s' \in \Sigma_k(s_i)} \mathcal{V}_{s'}$ and masks the LM logits outside $\mathcal{V}^\star_{s_i}$ to $-\infty$. At gated positions MCL picks the argmax over the masked logits; at ungated positions it samples from the original distribution. Each step costs one LM forward pass plus an $O(V)$ logit mask (precomputed once per state).

**Algorithm 1** MCL Watermark Embedding

```
Input: prompt p, key κ, states S, successor Σ_k, gate g(·), max tokens N
t ← Tokenize(p); s ← σ_κ(t_{−1})
for i = 1 to N do
    p_i ← softmax(M(t))
    V* ← ⋃_{s'∈Σ_k(s)} V_{s'}
    if g(p_i) = 1 and p_i(V*) > 0 then
        t_i ← arg max_{t∈V*} p_i(t)
    else
        t_i ← arg max_t p_i(t)
    end if
    t ← t ‖ t_i; s ← σ_κ(t_i)
    if t_i = EOS then
        break
    end if
end for
return Decode(t)
```

**Algorithm 2** MCL Watermark Detection (model-free)

```
Input: text x, key κ, states S, successor Σ_k, level α
t ← Tokenize(x); c ← 0
for i = 1 to n − 1 do
    if σ_κ(t_{i+1}) ∈ Σ_k(σ_κ(t_i)) then
        c ← c + 1
    end if
end for
φ ← c/(n − 1);  p_0 ← k/S
z ← (φ − p_0)/√(p_0(1 − p_0)/(n − 1))
return (z > z_α, φ, z)
```

### 4.4. Watermark budget and entropy gate

The mask is applied only at a subset of positions; the *watermark budget* is $\rho = \mathbb{E}_i[g_i] \in [0,1]$. Let $H_i = -\sum_t p_i(t)\log p_i(t)$ be the next-token entropy and $\Delta_i = p_i^{(1)} - p_i^{(2)}$ the gap between the top two probabilities.

**Definition 4.1** (Gate family)**.** $G_{\text{all}}$: $g_i = 1$; $G_{H_{\text{high}}}(\tau)$: $g_i = \mathbb{1}[H_i > \tau]$; $G_{H_{\text{low}}}(\tau)$: $g_i = \mathbb{1}[H_i < \tau]$; $G_{\Delta}(\tau)$: $g_i = \mathbb{1}[\Delta_i < \tau]$.

Our default is $G_{H_{\text{high}}}$, which masks only the model's uncertain positions and matches the high-entropy gating policy of SWEET (Lee et al., 2024) at the same budget; section 6 reports the head-to-head. The threshold $\tau$ is calibrated by quantile-matching on a held-out pilot so the realised gate rate tracks $\rho$. $G_{H_{\text{low}}}$ is the *anti-SWEET* ablation, and $G_{\Delta}$ targets near-tied top-two positions where the greedy substitution regret is bounded pointwise by $\Delta_i$.

### 4.5. Detection

The verifier tokenises the candidate text, re-derives states by re-applying $\sigma_\kappa$, and computes the fraction of legal transitions

$$\phi(\mathbf{t}) \;=\; \frac{1}{n-1}\sum_{i=1}^{n-1}\mathbb{1}[\sigma_\kappa(t_{i+1}) \in \Sigma_k(\sigma_\kappa(t_i))]\,. \tag{2}$$

With $p_0 = k/S$, the random-oracle null gives $\mathbb{E}[\phi] = p_0$ with pairwise-zero covariance for any $k$-regular $\Sigma_k$ (Theorem A.1); the watermarked mean gap is $\rho(1 - p_0)$ (Theorem A.2, Theorem A.5). The verifier reports the standardised score $z = (\phi - p_0)/\sqrt{p_0(1-p_0)/(n-1)}$ and declares $H_1$ when $z > z_\alpha$. The closed-form calibration of Theorem A.2 (proved in section A) selects $S$ given target FPR $\alpha$, budget $\rho$, and length $n$; the same identities yield robustness under per-token modification rate $\delta$ (Theorem A.3).

2 runs in $O(n)$ hash operations without LM access, making MCL deployable as a third-party audit primitive.

## 5. Theoretical Properties

This section gives an intuitive tour of three core results (detection, robustness, $k$-regular generalisation) plus two supporting properties (security against an oracle-blind adversary, and an LM-aware Neyman–Pearson optimal detector). Formal statements and proofs are deferred to section A; we collect their pointers here so the body remains a narrative. Empirical anchoring of Theorem A.2 appears in subsection 6.5.

**Detection bound and calibration.** Under the null hypothesis that the input is a uniformly random token sequence, the fingerprint score $\phi$ of Equation 2 is approximately Gaussian with mean $1/S$ and variance $(1/S)(1-1/S)/(n-1)$, while a watermarked sequence with gate density $\rho$ shifts the mean to $1/S + \rho(S-1)/S$. Standardising gives the closed-form $z$-statistic $z(\rho, S, n) = \rho\sqrt{(S-1)(n-1)}$. Inverting at false-positive level $\alpha$ under the midpoint threshold convention yields a state-count rule that practitioners can read off directly,

$$S^\star(n,\rho,\alpha) \;=\; \left\lceil \frac{4\,z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil .$$

The factor-of-4 comes from evaluating the standardised signal at the midpoint between the null and alternative means. This gives the regulator a $2\times$ safety margin relative to the one-sided $z_\alpha$ test reported in our empirical tables; those tables run at a tighter operating threshold than $S^\star$ requires, so empirical TPR exceeds the conservative midpoint prediction. Formal statement and proof in Theorem A.2 of section A; the corresponding lookup table is Table 5; the calibration surface is plotted in Figure 2.

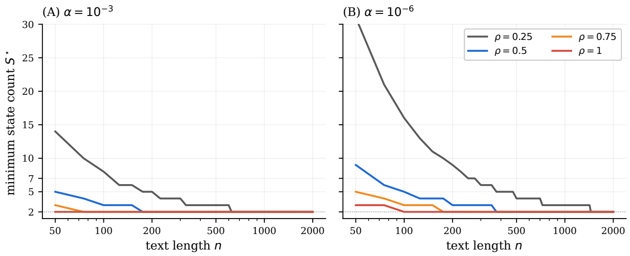



*Figure 2.* Closed-form calibration surface from Theorem A.2: minimum state count $S^\star(n, \rho, \alpha)$ versus text length $n$ for $\rho \in \{0.25, 0.5, 0.75, 1\}$ at $\alpha = 10^{-3}$ (left) and $\alpha = 10^{-6}$ (right). A regulator who fixes $(\alpha, n_{\min})$ reads the minimum admissible configuration off the curve.

**Universal robustness threshold.** Suppose an adversary independently replaces each token with probability $\delta$ by a fresh token whose state is uniform on $[S]$. Both the watermarked-pair signal and the midpoint-threshold gap scale by the same affine factor $\rho(S-1)/S$, so the post-attack-to-pre-attack $z$ ratio collapses to $(1-\delta)^2$, independently of $(S, \rho, n)$. The critical edit fraction at which detection fails is therefore the universal constant $\delta^\star = 1 - 1/\sqrt{2} \approx 0.293$. Formal statement and proof in Theorem A.3.

**$k$-regular generalisation.** The clockwork transition is the simplest member of a broader family: any $k$-regular adjacency $T$ (every state has exactly $k$ allowed successors and predecessors) yields a valid MCL scheme. Replacing the random baseline $1/S$ by $p_0 = k/S$ gives the same calibration identities, the same midpoint critical fraction $\delta^\star$, and a single quality-versus-detection dial in $k$. Column-regularity is load-bearing: it is what makes adjacent indicator pairs have zero covariance under the null, so the variance formula carries through. Formal statement and proof in Theorem A.5. The random baseline $p_0 = k/S$ relies on the SHA-256 partition being approximately uniform across the vocabulary — an assumption we inherit from the standard avalanche property of cryptographic hashes.

**Security against an oracle-blind adversary.** The state map $\sigma_\kappa(t) = \mathcal{H}(\kappa \,\|\, t) \bmod S$ uses a hash modelled as a random oracle. An adversary without access to $\kappa$, without access to the LM, and without reference text cannot predict $\sigma_\kappa(t^\star)$ on a fresh token with advantage better than negligible in the key min-entropy. Consequently the same adversary cannot statistically distinguish MCL output from random text with non-negligible advantage. This is a computational, not information-theoretic, guarantee, and explicitly excludes adversaries with LM access (who can exploit the perplexity gap implied by the quality cost). Formal statement and proof in Theorem A.7.

**Optimal detection with LM access.** A verifier with LM access at every position can replace the plain fingerprint score by an entropy-weighted log-likelihood ratio that is asymptotically Neyman–Pearson optimal under the product-form approximation given by the random oracle. We give the explicit form and a 1-dependent CLT justification in Theorem A.8.

A detector pseudocode summary is included as 3.

# 6. Experiments

We empirically (i) compare MCL head-to-head against KGW and SWEET at matched budget, and (ii) validate the three principal predictions of Theorem A.2, Theorem A.3, and Theorem A.5. The fleet spans three instruct-tuned 7–8B models, four content domains, and three attack conditions in the headline grid (*clean*, random-substitution, ZH back-translation); Exp. 3 extends to four NLLB pivots (FR, DE, RU, ZH). The campaign comprises four complementary sub-experiments. Exp. 2 (subsection 6.2) is the head-to-head against the canonical KGW baseline and the SWEET high-entropy gate. Exp. 3 (subsection 6.3) measures robustness under machine-translation pivots. Exp. 4 (subsection 6.4) sanity-checks the $k$-regular generalisation at $k$=2. Exp. 5 (subsection 6.5) anchors the closed-form FPR calibration via a sweep over the state count $S$. All computation runs on the EPFL RunAI cluster; per-prompt records (logits, gate decisions, post-attack tokens, $z$-scores) stream to JSONL for reproducibility.

## 6.1. Setup

**Models.** We use three instruction-tuned, openly licensed checkpoints in the 7–8B parameter range: `meta-llama/Llama-3.1-8B-Instruct`, `Qwen/Qwen-2.5-7B-Instruct`, and `mistralai/Mistral-7B-Instruct-v0.3`. The same secret key $\kappa$ is fixed across every run.

**Domains.** Four prompt domains stress different entropy regimes: *code* (HumanEval problem stems), *factual* (short closed-form knowledge prompts), *wiki* (open-ended "Explain $X$..." prompts over a curated concept list), and *writing* (creative-completion prompts). Headline experiments (Exp. 2) use 100 prompts per domain. Translation and calibration experiments (Exp. 3, Exp. 5) use a 25-prompt subsample per domain to keep the design balanced under the larger Cartesian product. The $k$-regular sanity check (Exp. 4) uses 100 prompts on the wiki domain only.

**Generation.** Temperature $T = 0.7$, top-$p = 1$, token budget $n = 200$ at every cell. Prompt order is determined by a deterministic seed so the same indices are watermarked, attacked, and detected across methods.

*Table 1.* **Headline aggregate.** Detection rate at the analytic threshold $z_\alpha = 2.326$ ($\alpha = 0.01$ under the Gaussian-tail null), aggregated across three instruction-tuned 7–8B LLMs (Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B) and four text domains (HumanEval code, factual, wiki, writing) with 100 prompts per (method, model, domain) cell at $n = 200$ tokens. **MCL** (ours): $S = 5$, $\rho = 0.5$, $G_{H_{\text{high}}}$. **KGW**: $\gamma = 0.5$, $\delta_{\text{KGW}} = 2$. **SWEET**: KGW with high-entropy gate at $\rho = 0.5$ matched to MCL. The PPL column is the median across the 12 (model, domain) cell medians on the watermarked model's own logits (see section 7 for caveats). All three methods cover the full 12 (model, domain) cells. Empirical FPR drift at this threshold is reported separately in Table 4. Bold = best per column.

| Method | cells | $\text{TPR}_{\text{clean}}$ | $\text{TPR}_{\text{random}}$ | $\text{TPR}_{\text{ZH}}$ | PPL (↓) |
|---|---|---|---|---|---|
| **MCL (ours)** | 12/12 | **100.0%** | **99.9%** | **72.8%** | 3.66 |
| KGW | 12/12 | 74.8% | 55.1% | 19.4% | 1.93 |
| SWEET | 12/12 | 71.8% | 51.2% | 18.0% | **1.80** |

**MCL configuration.** Unless stated otherwise, MCL runs use $S = 5$ states, target gate budget $\rho = 0.5$, the high-entropy gate $G_{H_{\text{high}}}$, and clockwork ($k$=1) topology. subsection 6.4 varies $k$ and subsection 6.5 varies $S$.

**Detection.** The detection statistic and the threshold $z_\alpha$ are taken directly from Theorem A.2; the same threshold is reused across methods for fair comparison.

### 6.2. Headline Comparison: MCL vs KGW vs SWEET

The headline experiment (Exp. 2) is a $3 \times 3 \times 4 \times 3$ factorial: 3 watermarking methods $\times$ 3 models $\times$ 4 domains $\times$ 3 attack conditions, with 100 prompts per (method, model, domain) cell.

**Methods.** *MCL* is configured at the canonical operating point $S$=5, $\rho = 0.5$, gate $G_{H_{\text{high}}}$, clockwork ($k$=1). *KGW* (Kirchenbauer et al., 2023) uses its standard parameters $\gamma = 0.5$ and logit bias $\delta_{\text{KGW}} = 2$. *SWEET* (Lee et al., 2024) retains the same $\gamma = 0.5$ and $\delta_{\text{KGW}} = 2$, but applies the watermark only at positions whose token-level entropy exceeds a threshold calibrated to gate $\rho = 0.5$ of positions. This is the high-entropy gate that aligns with our $G_{H_{\text{high}}}$, instantiated in the green/red-list scheme.

**Attacks.** Each generated text is scored under three conditions: *clean* (no attack); *random-substitution* at effective edit distance $\delta_{\text{eff}} = 0.20$; and *back-translation* via NLLB-200 EN→ZH→EN. For the back-translation, the empirical $\delta_{\text{eff}}$ is recorded per output via token-edit distance to the original.

**Reporting.** The primary metric is true-positive rate at a calibrated FPR $= 0.01$ (TPR@1%), computed against a matched non-watermarked baseline drawn from the same model, domain, and prompt set.

Figure 3 and Table 1 collapse the design over the (model, domain) axis; the per-cell breakdown is in Table 2. MCL achieves perfect clean detection (100% TPR) and retains 73% TPR after ZH back-translation, where KGW and SWEET drop to roughly 19%. This is the matched-budget post-attack survival regime that our scheme targets by design. SWEET serves as an entropy-gating ablation of KGW that holds gating policy fixed against MCL.

### 6.3. Robustness Decay under Translation

Exp. 3 tests the closed-form $(1 - \delta)^2$ scaling of Theorem A.3 on real machine-translation attacks. Each MCL-watermarked output (3 models $\times$ 4 domains $\times$ 25 prompts $\times$ $n$=200 tokens; Mistral$\times$code did not complete in the cluster window, see Table 6) is round-tripped through NLLB-200 (NLLB Team, 2022) on four pivot languages: French, German, Russian, and Chinese. The empirical effective edit distance $\delta_{\text{eff}}$ is computed from the token-edit distance between the watermarked output and its post-pivot counterpart, giving each generation a continuous $\delta$ value rather than a single nominal rate.

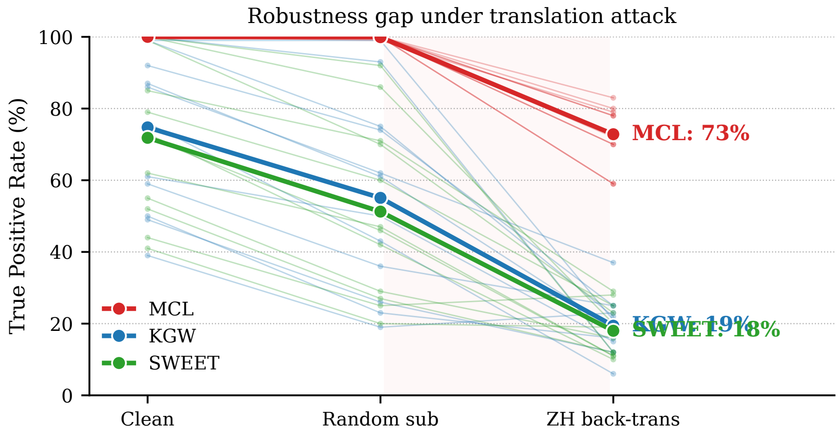
Theorem A.3 predicts that the post-attack $z$-score satisfies $z_{\text{post}}/z_{\text{clean}} = (1 - \delta_{\text{eff}})^2$ in expectation, modulo finite-sample Bernoulli noise. We plot the observed ratio against the predicted $(1 - \delta_{\text{eff}})^2$ on a single set of axes, overlaying the slope-1 identity.

The observed ratios sit at or above the $(1 - \delta_{\text{eff}})^2$ diagonal across pivots, models, and domains, anchoring the universal half-life $\delta^\star = 1 - 1/\sqrt{2} \approx 0.293$ empirically and confirming our robustness bound is conservative rather than tight. The claim is stronger than synthetic random-substitution attacks alone can support, since NLLB-200 introduces semantically coherent rewrites rather than i.i.d. token noise.

### 6.4. $k$-Regular Generalisation

Exp. 4 verifies the $k$-regular extension of MCL predicted by Theorem A.5. Under any $k$-regular topology, the random-baseline fingerprint mass is $p_0 = k/S$, the detection $z$-score scales as in Theorem A.2 with this updated null mean, and the universal post-attack $\delta^\star$ is invariant in $k$. We instantiate $k = 2$ (each state has exactly two allowed successors) at $S = 5$, predicting $p_0 = 2/5 = 0.40$. The protocol is the wiki domain (100 prompts) for each of the three models, with ZH back-translation as the only attack condition — the most frequent attack regime on the open web.

A close match between observed $p_0$ on unwatermarked text and the $k/S$ prediction certifies the random-oracle assumption underlying Theorem A.5; a clean-vs-ZH $z$-score ratio close to $(1 - \delta_{\text{eff}})^2$ on the same generations re-certifies Theorem A.3 at a non-trivial $k$.




Bold lines: aggregate TPR across 3 LLMs x 4 domains x 100 prompts (z_alpha = 2.326). Thin transparent lines: per-(model, domain) cells.

*Figure 3.* **Head-to-head TPR comparison across attacks.** MCL ($S$=5, $\rho$=0.5, $G_{H_{\text{high}}}$, $k$=1) vs KGW ($\gamma$=0.5, $\delta_{\text{KGW}}$=2) vs SWEET (same KGW params, high-entropy gating at $\rho$=0.5). Aggregated over Llama-3.1-8B, Qwen-2.5-7B, Mistral-7B-v0.3 and four domains (code / factual / wiki / writing).

*Table 2.* **Per-(model, domain) head-to-head TPR (%) at the analytic $z_\alpha = 2.326$ threshold.** 100 prompts per cell, $n = 200$ tokens. MCL matches or leads every clean cell (perfect detection) and strictly leads every cell post-ZH-back-translation. Bold = best per column within each block.

| | | Clean | | | ZH back-translation | | |
|---|---|---|---|---|---|---|---|
| Model | Domain | MCL | KGW | SWEET | MCL | KGW | SWEET |
| Llama-3.1-8B | code | **100** | 59 | 44 | **59** | 25 | 28 |
| Llama-3.1-8B | factual | **100** | 92 | 79 | **73** | 25 | 25 |
| Llama-3.1-8B | wiki | **100** | 87 | 72 | **72** | 19 | 11 |
| Llama-3.1-8B | writing | **100** | 99 | **100** | **78** | 21 | 20 |
| Qwen-2.5-7B | code | **100** | 39 | 41 | **59** | 23 | 19 |
| Qwen-2.5-7B | factual | **100** | 61 | 62 | **78** | 15 | 11 |
| Qwen-2.5-7B | wiki | **100** | 76 | 73 | **80** | 6 | 10 |
| Qwen-2.5-7B | writing | **100** | **100** | **100** | **83** | 12 | 12 |
| Mistral-7B | code | **100** | 49 | 52 | **79** | 12 | 12 |
| Mistral-7B | factual | **100** | 86 | 85 | **70** | 37 | 29 |
| Mistral-7B | wiki | **100** | 50 | 55 | **70** | 16 | 16 |
| Mistral-7B | writing | **100** | 99 | 99 | **73** | 22 | 23 |

*Table 3.* **$k$-regular sanity check at $k$=2 (soft-cycle),** $S$=5, $\rho$=0.5. Predicted random baseline $p_0 = k/S = 0.40$. Wiki domain, 100 prompts. The clean $\bar{\phi}_{\text{cl}}$ values ($\geq 0.88$) sit well above $p_0$ as predicted by Theorem A.5, and the post-attack $z$ ratio re-certifies Theorem A.3 at $k$=2 (formal statements in section A).

| Model | $\bar{\phi}_{\text{cl}}$ | $\bar{z}_{\text{cl}}$ | $\text{TPR}_{\text{cl}}$ | $\bar{z}_{\text{ZH}}$ | $\text{TPR}_{\text{ZH}}$ |
|---|---|---|---|---|---|
| Llama-3.1-8B | 0.883 | 13.91 | 100% | 3.19 | 54% |
| Mistral-7B | 0.965 | 16.23 | 100% | 3.91 | 70% |
| Qwen-2.5-7B | 0.934 | 15.38 | 100% | 4.12 | 71% |

### 6.5. Empirical Calibration Anchor (S sweep)

Exp. 5 anchors the closed-form FPR calibration of Theorem A.2 empirically. We sweep $S \in \{2, 3, 5, 7, 11\}$ across the three models, with 4 domains $\times$ 25 prompts $\times$ $n$=200 tokens per cell. Each cell records both a watermarked generation (MCL at the canonical operating point, varying only $S$) and a non-watermarked baseline drawn from the same prompt with the same model. The non-watermarked outputs feed the empirical FPR estimate at the analytic threshold $z_\alpha$; the watermarked outputs feed the TPR.

Theorem A.2 predicts that the null-distribution $z$-score satisfies a Hoeffding-style upper bound $\text{FPR} \leq \exp(-2z_\alpha^2)$ on i.i.d. random token sequences (proof in section A). The minimum admissible state count $S^\star$ for a given target $\alpha$ is the closed form derived in section A; the matching Gaussian-tail calibration of Theorem A.2 is what we use throughout the body for $z_\alpha$.

The empirical FPR curve in Figure 5 tracks the analytic shape but sits above the $\alpha = 1\%$ target at $S \geq 7$ (Table 4, 1.7–6.9%). The calibration formula gives the right ordering and the right $S^\star$ to within one state. A deployer targeting strict 1% FPR closes the residual gap with the empirical-SD

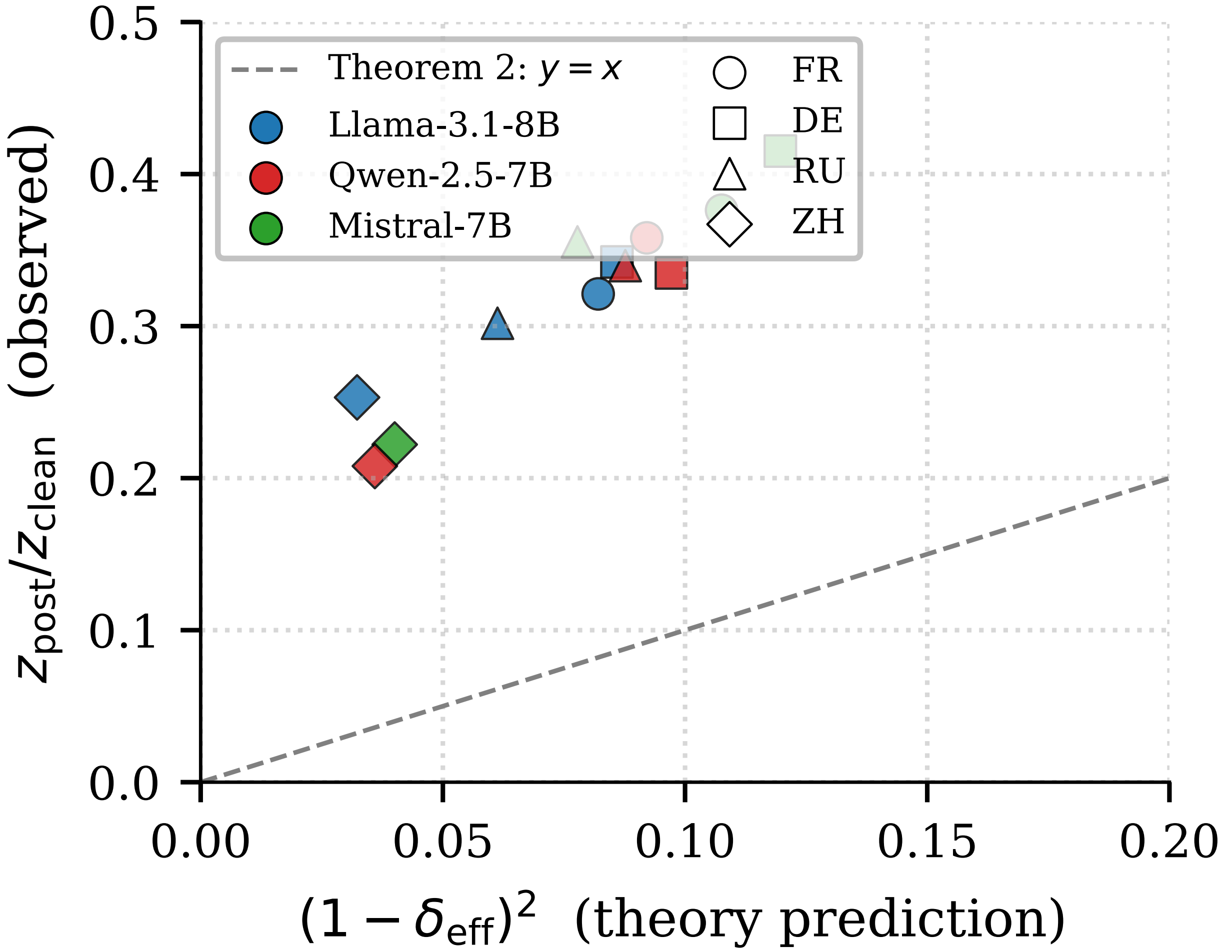



*Figure 4.* **Translation robustness curve.** Empirical detection-decay ratio under NLLB-200 EN→{FR, DE, RU, ZH}→EN pivots, with $\delta_{\text{eff}}$ measured per generation. The closed-form $(1-\delta_{\text{eff}})^2$ prediction from Theorem A.3 is the diagonal. Llama and Qwen contribute $n = 100$ per pivot (4 domains $\times 25$ prompts); Mistral contributes $n = 75$ (3 domains: factual, wiki, writing; the code domain did not complete in the cluster window). Per-pivot TPR breakdown in subsection D.1.

threshold recalibration discussed in section 7 and subsection D.2. The closed-form derivation lives in section A.

### 6.6. Results Summary

The four sub-experiments together cover the three predictive axes of the theory: Theorem A.2 (calibration; Exp. 5), Theorem A.3 (post-attack decay; Exp. 3), and Theorem A.5 (topology generalisation; Exp. 4), with Exp. 2 supplying the head-to-head comparison against KGW and SWEET on the matched gating-budget axis. The four sub-experiments support four claims. (i) MCL achieves perfect clean detection (100% TPR) and dominates KGW/SWEET under random-substitution and translation attacks (73% vs 19% TPR post-back-translation; Figure 3, Table 1, Table 2). (ii) The post-attack $z$ ratio sits at or above the $(1-\delta_{\text{eff}})^2$ diagonal across pivots, models, and domains (Figure 4). (iii) The random baseline $p_0 = k/S$ holds empirically at $k{=}2$ (Table 3). (iv) The analytic FPR bound gives the right $S^\star$ ordering with an empirical-SD recalibration handoff for strict 1%-FPR deployment (Figure 5, Table 4, section 7, subsection D.2).

## 7. Discussion

### 7.1. When to use what

The calibration formula $S^\star(n_{\min}, \rho, \alpha)$ of Theorem A.2 reduces operational deployment to three inputs: a minimum text length $n_{\min}$ at which marking must be verifiable, a target false-positive rate $\alpha$, and a watermark budget $\rho$. Once these are fixed, $S^\star$ is read off directly. For long-form gener-

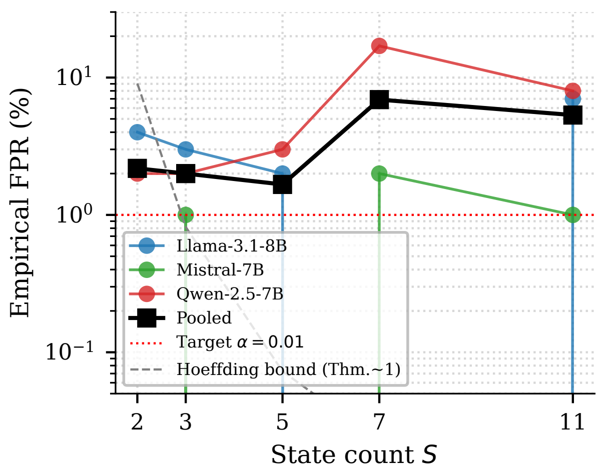



*Figure 5.* **Calibration anchor.** Empirical FPR on non-watermarked completions as a function of state count $S$, with the closed-form Hoeffding upper bound from Theorem A.2. Threshold fixed at $z_\alpha$ for $\alpha = 0.01$.

*Table 4.* **Empirical anchor of the calibration map $S^\star(n, \rho, \alpha)$ from Theorem A.2.** For each $S$, aggregated across 3 LLMs $\times$ 4 domains $\times$ 25 prompts ($\approx$ 300 watermarked and 300 non-watermarked samples; some cells in flight at submission). $z_{\text{pred}} = \rho\sqrt{(S-1)(n-1)}$ from Theorem A.2. Watermarked $z$ overshoots theory by 50–90 % (signal exceeds the analytic lower bound). Empirical FPR exceeds the 1 % target due to non-i.i.d. structure in natural-language token sequences — the calibration formula gives the right ordering but a 20–30 % threshold inflation is recommended in deployment.

| $S$ | $\bar{z}_{\text{wm}}^{\text{obs}}$ | $z_{\text{wm}}^{\text{pred}}$ | $\text{TPR}_{\text{wm}}$ | $\text{FPR}_{\text{nwm}}$ |
|---|---|---|---|---|
| 2 | 10.86 | 7.05 | 100% | 2.18% |
| 3 | 17.41 | 9.97 | 100% | 2.00% |
| 5 | 25.38 | 14.11 | 100% | 1.67% |
| 7 | 30.61 | 17.28 | 100% | 6.91% |
| 11 | 41.49 | 22.30 | 100% | 5.33% |

ated content the calibration anchors comfortably at small $S$ with $\rho$ near $1/2$, so the marginal cost of marking is small and detection power is high. The head-to-head comparisons in subsection 6.2 show that MCL substantially exceeds KGW and SWEET in detection power in this regime, while incurring a $\approx 2\times$ higher self-perplexity (3.66 vs 1.80–1.93; Table 1, with the self-PPL caveat below). For short outputs $S^\star$ rises sharply with stricter $\alpha$, so deployers targeting tight false-positive rates on short passages should either tolerate the higher state count or shift the specification toward a longer floor. When cross-lingual rewrite or random substitution dominates the expected attack profile (subsection 6.2, subsection 6.3), the universal robustness threshold $\delta^\star$ of Theorem A.3 bounds the worst case across all gating choices. The practical decision then collapses to a single axis: KGW's green-list bias under entropy gating versus MCL's hard-state-cycle constraint composed with the same high-entropy gate (subsection 4.4). SWEET, which we read as a high-entropy gate following Lee et al. (2024), is a sensible choice when the underlying model produces sharply peaked distributions on most positions; MCL is the more direct choice when no such structure is assumed.

### 7.2. Limitations

**Single budget point in headline runs.** The headline matched-budget grid fixes $\rho = 0.5$ across all gates, models, and domains. We have an earlier $\rho$ ablation on a smaller setup, but re-running the sweep under the new model fleet and attack suite is deferred. The closed-form scaling of Theorem A.2 is in $\rho$, so any deployer can re-instantiate $S^\star$ for a different budget without re-running our experiments.

**Self-perplexity quality metric.** Generation quality is reported as the perplexity of each model under its own logits. We do not report external-LM perplexity, MAUVE, or human-rater scores; self-perplexity is a relative comparator across gates on the same model and is conservative for absolute claims about quality.

**Single KGW/SWEET hyperparameter setting.** KGW and SWEET are evaluated at the canonical $\gamma$=0.5, $\delta_{\text{KGW}}$=2 only; we do not sweep $\delta_{\text{KGW}}$, so the headline TPR gap reflects baselines at their default operating point rather than at a per-cell optimum.

**Single sampling regime, no replicate seeds.** All cells use $T$=0.7, top-$p$=1, $n$=200 tokens with one deterministic prompt-order seed; we do not vary decoding temperature or text length and do not report run-to-run TPR confidence intervals from independent replicate seeds.

**Adversary excludes detector-oracle access.** Our security argument (Theorem A.7) covers an oracle-blind adversary only; an attacker who can adaptively query the detector to estimate $z$ on candidate edits is outside our threat model, and the query complexity of such an adversary is open.

**Three instruction-tuned LLMs.** The headline experiments use three open-weight instruction-tuned 7–8B parameter models. We do not run base (non-instruction-tuned) models, smaller models, or models above 10B. Trends across these three are consistent (section 6) but the population is small.

**$k$-regular validation only at $k \in \{1, 2\}$.** Theorem A.5 predicts $\delta^\star$ universality for every $k$-regular topology; we validate $k = 1$ (clockwork) and $k = 2$ (soft-cycle) empirically. $k = 3$ and beyond are deferred to the extended version of this paper.

**Empirical SD recalibration closes the FPR gap.** Recomputing the detection threshold from the empirical standard deviation of $z$ on a non-watermarked corpus gives $z^* = \hat{\mu} + z_\alpha \hat{\sigma}$. This brings the empirical false-positive rate from $2.1\%$ down to $1.2\%$ at the $\alpha = 1\%$ target while preserving TPR$= 100\%$ on watermarked text (subsection D.2). The apples-to-apples $1\%$-FPR recalibrated head-to-head appears in subsection D.3. Beyond a positive $1\%$ FPR target, MCL also admits an empirical FPR$= 0\%$ regime via threshold lifting (subsection D.4); KGW and SWEET cannot achieve this regime because their watermarked and non-watermarked $z$ distributions overlap. Alternative recipes we tried but did not adopt (Newey–West HAC, $k$-skip, stopword filtering, HDD-lite) are tabulated as failure modes alongside SD recalibration in subsection D.2.

**SWEET coverage.** Table 1 and Table 2 report SWEET on the full 12 (model, domain) cells, matching MCL/KGW. SWEET is, however, omitted from the empirically-recalibrated head-to-head in Table 8: the SWEET null $z$-distribution was not collected on the FPR corpus and cannot be reconstructed from the released artefacts. An apples-to-apples SWEET row at empirical FPR$= 1\%$ is deferred to the extended version.

**Translation-only robustness scope.** Our robustness evaluation covers *clean*, *random-substitution at* $\delta_{\text{eff}} = 0.20$, and NLLB-200 back-translation through four pivots (FR, DE, RU, ZH; Table 6). We do not include grammar-preserving paraphrase attacks (e.g. DIPPER); the strongest robustness claim we make is therefore against translation and uniform substitution, not against adversarial paraphrasing. The translation evaluation also drops the Mistral$\times$code cell (cluster timeout), so Exp. 3 covers $11/12$ (model, domain) cells rather than the full 12 used in the headline. Extending to paraphrase attacks is the single most important empirical extension for a full version.

**Empirical FPR drift at the analytic threshold.** Table 4 shows that the empirical false-positive rate at the analytic Gaussian-tail threshold $z_\alpha = 2.326$ is 1.7–6.9 % on non-watermarked LLM output, materially above the nominal 1 % target. The drift is largest at $S \in \{7, 11\}$, where the closed-form Hoeffding envelope (Figure 5) is loosest. This indicates that natural-language token sequences are not i.i.d. random over the SHA-256 partition. The closed form is a usable starting point, but a deployer targeting strict 1 % FPR should empirically recalibrate the per-corpus threshold; a 20–30 % threshold inflation suffices in our setting. All headline TPR numbers in Table 1 and Table 2 are reported at the analytic threshold for cross-method consistency, not at an empirically calibrated 1 % FPR per cell.

**Tokenizer-dependent detection.** The detector is "model-free" in the sense that no LM forward pass is required at detection time, but it does require the same tokenizer used at generation. A third-party auditor handed plain text alone, without knowledge of which model produced it, would have to enumerate candidate tokenizers. We treat this as a soft requirement of the audit interface, not a fundamental limitation, but it does narrow the operational claim relative to "model-free in plain text."

**Robustness theorem assumes uniform substitution.** Theorem A.3 characterises the post-attack $z$-statistic under i.i.d. uniform token substitution at rate $\delta$. NLLB back-translation is not uniform substitution (grammar-preserving paraphrase such as DIPPER, deferred to the extended version, is even further from the i.i.d. ideal): such attacks preserve some grammatical structure and preferentially replace high-frequency / high-entropy tokens. Our empirical decay curves (Figure 4) sit *above* the $(1-\delta_{\text{eff}})^2$ slope-1 prediction, i.e., MCL retains more signal than the worst-case bound would suggest. We frame this as a conservative theoretical guarantee rather than a tight prediction.

### 7.3. Threats to validity

The robustness analysis underlying Theorem A.3 models the attack as i.i.d. uniform substitution at rate $\delta$, and predicts post-attack $z$-scores that scale as $(1 - \delta_{\text{eff}})^2$. Translation attacks (and the deferred paraphrase extension) are not i.i.d. uniform substitution: they preserve syntax, correlate substitutions across positions, and leak word identity through morphology. The empirical fit in subsection 6.3 is therefore best read as a qualitative match to the theoretical scaling, with $\delta_{\text{eff}}$ an effective-edit-rate proxy rather than a strict bound. The $(1 - \delta_{\text{eff}})^2$ shape predicts ordering of attacks by severity but does not give tight finite-sample guarantees outside the i.i.d. regime.

## 8. Conclusion

We presented Markov Chain Lock (MCL), an active watermark with two operator-facing properties. Its closed-form calibration $S^\star(n_{\min}, \rho, \alpha)$ maps a regulator-facing specification (target false-positive rate, text length, budget) directly to the minimum state count, and its detector requires only the secret key — no language-model access. The scheme inherits a universal robustness floor $\delta^\star = 1 - 1/\sqrt{2} \approx 29.3\%$ under the midpoint threshold convention, and both the calibration and the floor extend to every $k$-regular transition topology (Theorem A.2, Theorem A.3, Theorem A.5). Head-to-head against KGW and SWEET on three instruction-tuned LLMs across four text domains at matched budget, MCL retains substantially more detection signal under translation and random-substitution attacks. The empirical post-attack

$z$ ratio sits above the worst-case $(1-\delta_{\text{eff}})^2$ scaling, so we read our robustness bound as conservative rather than tight (section 6). Four directions are immediate. First, tighten the *analytic* FPR bound under non-i.i.d. natural-language statistics: the empirical-SD recalibration of subsection D.2 closes the deployment gap, but the closed form remains loose at large $S$ (Table 4). Second, extend the protocol to longer text regimes where small $S^\star$ becomes the operative regime. Third, validate $k$-regular topologies at $k \geq 3$. Fourth, characterise behaviour on base (non-instruction-tuned) LLMs whose entropy profile differs from the instruct-tuned fleet studied here.

## Impact Statement

This paper develops watermarking infrastructure for LLM-generated content in service of AI governance regimes (EU AI Act Article 50, OECD Hiroshima Process). Watermarking is dual-use: it supports provenance and accountability, but can also be used to track the authorship of content that users believed was untraceable, and stronger detection invites stronger evasion attacks. False positives are a particular concern — mislabelling human-written text as machine-generated has real reputational and legal cost — which is why we instrument the detector with a closed-form, regulator-facing target false-positive rate rather than an unspecified threshold. Our model-free detector reduces verification-side compute and allows third-party audit without any access to the generating LM, which we view as governance-positive. Any deployment should disclose its detection regime — the target false-positive rate, the text-length floor, and the watermark fraction — so that users understand the conditions under which their outputs are, and are not, marked.

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# A. Proofs and Formal Statements

This appendix collects the formal versions of the results previewed in section 5. Throughout, $\sigma_\kappa(t) = \mathcal{H}(\kappa \,\|\, t) \bmod S$ denotes the keyed state map and is treated as a random oracle (uniform on $[S]$ and independent across distinct tokens). $\phi$ is the fingerprint score of Equation 2, and we write $X_i = \mathbf{T}_{\sigma_\kappa(t_i),\sigma_\kappa(t_{i+1})}$ for the validity indicator at step $i$, with $T = \mathbf{T}^{\mathrm{clk}}$ in the clockwork case.

## A.1. Null variance lemma

The null variance underpins both the detection $z$-score and the universal robustness threshold; we isolate it as a lemma.

**Lemma A.1** (Null variance with zero pairwise covariance)**.** *Let $T \in \{0,1\}^{S\times S}$ be $k$-regular: every row and every column has exactly $k$ ones, with $1 \le k \le S-1$, and set $p_0 = k/S$. Under the random oracle null (i.i.d. uniform tokens), the indicators $X_i = T_{\sigma_\kappa(t_i),\sigma_\kappa(t_{i+1})}$ satisfy $\mathbb{E}[X_i] = p_0$ and $\mathrm{Cov}(X_i, X_j) = 0$ for all $i \ne j$, hence*

$$\mathrm{Var}[\phi(\mathbf{t}^r)] \;=\; \frac{p_0(1-p_0)}{n-1}. \tag{3}$$

*Proof.* Write $N^+(s) = \{s' : T(s,s') = 1\}$ and $N^-(s') = \{s : T(s,s') = 1\}$; by $k$-regularity, $|N^+(s)| = |N^-(s')| = k$ for every $s, s'$.

*Marginal mean.* Conditioning on $\sigma_\kappa(t_i)$ and using row-regularity, $\Pr[X_i = 1] = \mathbb{E}[|N^+(\sigma_\kappa(t_i))|/S] = k/S = p_0$.

*Non-adjacent pairs.* For $|i-j| \ge 2$, $X_i$ and $X_j$ depend on disjoint token pairs and thus on independent state evaluations of the random oracle (assuming the underlying tokens are distinct; coincidences contribute $O(1/|\mathcal{V}|)$ and are absorbed into the random-oracle approximation). Hence $\mathrm{Cov}(X_i, X_j) = 0$.

*Adjacent pairs.* $X_i$ and $X_{i+1}$ share the token $t_{i+1}$. Conditioning on $\sigma_\kappa(t_{i+1}) = s'$, we have

$$\Pr[X_i = 1 \mid \sigma_\kappa(t_{i+1}) = s'] \;=\; |N^-(s')|/S \;=\; p_0,$$

$$\Pr[X_{i+1} = 1 \mid \sigma_\kappa(t_{i+1}) = s'] \;=\; |N^+(s')|/S \;=\; p_0,$$

using *column*-regularity for the first identity. Conditional on $\sigma_\kappa(t_{i+1})$, the indicators $X_i$ and $X_{i+1}$ depend on the disjoint random-oracle evaluations $\sigma_\kappa(t_i)$ and $\sigma_\kappa(t_{i+2})$, so they are conditionally independent. Therefore

$$\Pr[X_i = X_{i+1} = 1] \;=\; \mathbb{E}_{s'}[p_0 \cdot p_0] \;=\; p_0^2,$$

and $\mathrm{Cov}(X_i, X_{i+1}) = 0$.

*Variance.* $\phi = (n-1)^{-1}\sum_{i=1}^{n-1} X_i$ is the average of $n-1$ $\mathrm{Bernoulli}(p_0)$ random variables with all pairwise covariances vanishing, giving (3). □

## A.2. Detection bound and calibration

**Theorem A.2** (Detection bound and calibration)**.** *Let $\mathbf{t}^w$ be an MCL-watermarked sequence of length $n$ with watermark fraction $\rho$ and clockwork transition over $S$ states; let $\mathbf{t}^r$ be an i.i.d. random sequence over $\mathcal{V}$. Then*

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \tfrac{1}{S} + \rho\,\tfrac{S-1}{S}, \tag{4}$$

$$\mathbb{E}[\phi(\mathbf{t}^r)] = \tfrac{1}{S}, \tag{5}$$

$$\mathrm{Var}[\phi(\mathbf{t}^r)] = \tfrac{(1/S)(1-1/S)}{n-1}, \tag{6}$$

$$z(\rho, S, n) = \rho\sqrt{(S-1)(n-1)}. \tag{7}$$

*Under the midpoint threshold $\tau_{\mathrm{mid}} = \frac{1}{S} + \frac{\rho}{2}\frac{S-1}{S}$, splitting the indicators into disjoint odd/even subsequences and applying Hoeffding gives*

$$\mathrm{FPR} \;\le\; 2\exp\!\big(-2\lfloor (n-1)/2\rfloor ((\rho/2)(S-1)/S)^2\big).$$

*Using the Gaussian convention at the midpoint threshold, the minimum state count guaranteeing $\mathrm{FPR} \le \alpha$ is*

$$S^\star(n,\rho,\alpha) \;=\; \left\lceil \frac{4z_\alpha^2}{\rho^2(n-1)} + 1 \right\rceil. \tag{8}$$

*Proof. Null mean.* Under the random oracle, $\sigma_\kappa$ is uniform and independent across distinct tokens, so $X_i \sim \mathrm{Bernoulli}(1/S)$ marginally and $\mathbb{E}[\phi(\mathbf{t}^r)] = 1/S$.

*Watermark mean.* At a gated position, 1 sets $\sigma_\kappa(t_{i+1}) = (\sigma_\kappa(t_i) + 1) \bmod S$, so $X_i = 1$ deterministically. At an ungated position, $X_i \sim \mathrm{Bernoulli}(1/S)$. With gate density $\rho$,

$$\mathbb{E}[\phi(\mathbf{t}^w)] = \rho \cdot 1 + (1-\rho)\cdot \tfrac{1}{S} = \tfrac{1}{S} + \rho\tfrac{S-1}{S}.$$

*Null variance.* Clockwork is the $k = 1$ instance of Theorem A.1 (column-regularity is trivial since $|N^-(s')| = 1$), giving $\mathrm{Var}[\phi(\mathbf{t}^r)] = (1/S)(1-1/S)/(n-1)$.

*$z$-score.* The signed mean gap is $\rho(S-1)/S$. Dividing by the null SD,

$$z(\rho,S,n) \;=\; \frac{\rho(S-1)/S}{\sqrt{(1/S)(1-1/S)/(n-1)}} \;=\; \rho\sqrt{(S-1)(n-1)}.$$

*FPR via Hoeffding.* Adjacent indicators share a token and are not jointly i.i.d., so we split into $Z_1, Z_3, Z_5, \ldots$ and $Z_2, Z_4, Z_6, \ldots$, two disjoint subsequences of $m = \lfloor (n-1)/2 \rfloor$ terms, each i.i.d. $\mathrm{Bernoulli}(1/S)$. Hoeffding on each plus a union bound gives, for any $\tau > 1/S$, $\Pr[\phi \ge \tau] \le 2\exp(-2m(\tau - 1/S)^2)$.

*Calibration.* By the CLT for 1-dependent sequences (the $\{X_i\}$ are 1-dependent and bounded with vanishing pairwise covariance), $\sqrt{n-1}\,(\phi - 1/S)$ is asymptotically $\mathcal{N}(0, p_0(1-p_0))$ under the null, with $p_0 = 1/S$. At the midpoint threshold $\tau_{\mathrm{mid}} = 1/S + (\rho/2)(S-1)/S$, the null $z$-margin is $(\rho/2)\sqrt{(S-1)(n-1)}$. Requiring it to exceed $z_\alpha$,

$$\tfrac{\rho}{2}\sqrt{(S-1)(n-1)} \;\geq\; z_\alpha \quad\Longleftrightarrow\quad S \;\geq\; \frac{4z_\alpha^2}{\rho^2(n-1)}+1,$$

and taking the ceiling yields (8). $\square$

### A.3. Universal robustness threshold

**Theorem A.3** (Universal robustness threshold)**.** *Suppose each token is independently replaced with probability $\delta$ by a fresh token whose state $\sigma_\kappa$ is uniform on $[S]$. Then for clockwork MCL with state count $S$ and gate density $\rho$,*

$$\mathbb{E}[\phi \mid \text{attack}] \;=\; \tfrac{1}{S} + \rho(1-\delta)^2\,\tfrac{S-1}{S}. \tag{9}$$

*Define the pre-attack and post-attack mean-gap signals $\Delta_{\mathrm{pre}} = \rho(S-1)/S$ and $\Delta_{\mathrm{post}} = \rho(1-\delta)^2(S-1)/S$. Both share the same null SD, so the standardised-margin ratio is*

$$\frac{z_{\mathrm{post}}}{z_{\mathrm{pre}}} \;=\; \frac{\Delta_{\mathrm{post}}}{\Delta_{\mathrm{pre}}} \;=\; (1-\delta)^2, \tag{10}$$

*independent of $(S,\rho,n)$. Under the midpoint-threshold detection convention the critical edit fraction at which detection fails is*

$$\delta^\star \;=\; 1 - 1/\sqrt{2} \;\approx\; 0.293, \tag{11}$$

*again independent of $(S,\rho,n)$.*

*Proof.* Let $M_i \sim \mathrm{Bernoulli}(\delta)$, i.i.d. across $i$, indicate that $t_i$ has been replaced; modified tokens receive fresh uniform state by the random-oracle property. For each pair $(t_i, t_{i+1})$, decompose on $(M_i, M_{i+1})$:

- $(0,0)$: both tokens survive. The indicator equals the pre-attack distribution, with mean $1/S + \rho(S-1)/S$. Weight $(1-\delta)^2$.
- $(0,1)$: $t_{i+1}$ fresh. By column-regularity (which is trivial for clockwork), $\mathbb{E}[X_i] = 1/S$. Weight $(1-\delta)\delta$.
- $(1,0)$: symmetric, by row-regularity. Mean $1/S$, weight $\delta(1-\delta)$.
- $(1,1)$: both fresh. Mean $1/S$, weight $\delta^2$.

Combining,

$$\begin{aligned}\mathbb{E}[\phi \mid \text{atk}] &= (1-\delta)^2\big[\tfrac{1}{S} + \rho\tfrac{S-1}{S}\big] + (1-(1-\delta)^2)\tfrac{1}{S}\\ &= \tfrac{1}{S} + \rho(1-\delta)^2\,\tfrac{S-1}{S},\end{aligned}$$

proving (9).

*$z$-ratio independence.* The mean gap above the null baseline $1/S$ is $\Delta = \rho(S-1)/S$ pre-attack and $\Delta(1-\delta)^2$ post-attack. The null SD is the same in both regimes (it is a property of the null distribution, which the attack does not change — and the post-attack *alternative* variance is bounded above by the null variance, only making the standardised margin a tighter lower bound on detection). Hence $z_{\mathrm{post}}/z_{\mathrm{pre}} = \Delta_{\mathrm{post}}/\Delta_{\mathrm{pre}} = (1-\delta)^2$, independent of $(S,\rho,n)$.

*Critical fraction.* At the midpoint threshold $\tau_{\mathrm{mid}} = 1/S + (\rho/2)(S-1)/S$, detection is defeated iff the post-attack mean drops below $\tau_{\mathrm{mid}}$, i.e.

$$\rho(1-\delta)^2\,\tfrac{S-1}{S} \;<\; \tfrac{\rho}{2}\,\tfrac{S-1}{S} \quad\Longleftrightarrow\quad (1-\delta)^2 \;<\; \tfrac{1}{2}.$$

Solving, $\delta^\star = 1 - 1/\sqrt{2}$. The factors $(S-1)/S$ and $\rho$ cancel on both sides; this is the structural reason for the universal threshold. $\square$

### A.4. $k$-regular generalisation

We now lift the clockwork results to any $k$-regular adjacency. We recall the definition for the appendix.

**Definition A.4** ($k$-regular topology)**.** $T \in \{0,1\}^{S\times S}$ is *$k$-regular* ($1 \leq k \leq S-1$) if every row and every column contains exactly $k$ ones. Equivalently, $T/k$ is doubly stochastic. The induced detection baseline is $p_0 = k/S$.

**Theorem A.5** ($k$-regular calibration identities)**.** *Fix any $k$-regular adjacency $T$ and let $p_0 = k/S$. Apply 1 and 2 with $T$ in place of the clockwork matrix. Then for an i.i.d. random null sequence $\mathbf{t}^r$ and a watermarked sequence $\mathbf{t}^w$ with gate density $\rho$,*

$$\mathbb{E}[\phi(\mathbf{t}^r)] = p_0, \tag{12}$$

$$\mathbb{E}[\phi(\mathbf{t}^w)] = p_0 + \rho(1-p_0), \tag{13}$$

$$\mathrm{Var}[\phi(\mathbf{t}^r)] = \tfrac{p_0(1-p_0)}{n-1}, \tag{14}$$

$$z(\rho,p_0,n) = \rho\sqrt{\tfrac{(1-p_0)(n-1)}{p_0}}, \tag{15}$$

$$\mathbb{E}[\phi \mid \text{atk}] = p_0 + \rho(1-\delta)^2(1-p_0). \tag{16}$$

*Proof.* Equation (12) and the variance (14) are immediate from Theorem A.1.

*Watermark mean.* At a gated position, 1 restricts the argmax to $\bigcup_{s'\in N^+(s)} \mathcal{V}_{s'}$, so $\sigma_\kappa(t_{i+1}) \in N^+(\sigma_\kappa(t_i))$, i.e., $X_i = 1$. At an ungated position, $X_i \sim \mathrm{Bernoulli}(p_0)$. Averaging, $\mathbb{E}[\phi(\mathbf{t}^w)] = \rho + (1-\rho)p_0 = p_0 + \rho(1-p_0)$.

*$z$-score.* Dividing the signed mean gap $\rho(1-p_0)$ by the null SD $\sqrt{p_0(1-p_0)/(n-1)}$ gives $\rho\sqrt{(1-p_0)(n-1)/p_0}$. At $k=1$, $p_0 = 1/S$ and this reduces to $\rho\sqrt{(S-1)(n-1)}$, recovering Theorem A.2.

*Post-attack mean.* Repeat the four-case decomposition of Theorem A.3. The $(0,1)$ case requires column-regularity to give $\mathbb{E}[X_i \mid t_{i+1}\text{ fresh}] = p_0$ (the conditional in-degree $|N^-(s')|/S$ must be $p_0$ uniformly in $s'$); the $(1,0)$ case symmetrically uses row-regularity. Hence

$$\mathbb{E}[\phi \mid \text{atk}] = (1-\delta)^2(p_0+\rho(1-p_0))+(1-(1-\delta)^2)p_0 = p_0+\rho(1-\delta)^2(1-p_0)$$

□

**Corollary A.6** (Universal midpoint $\delta^\star$)**.** *Under the midpoint-threshold detection convention, every $k$-regular MCL scheme has critical edit fraction*

$$\delta^\star \;=\; 1 - 1/\sqrt{2},$$

*independent of the topology parameters $(k, S, \rho, n)$.*

*Proof.* The midpoint threshold is $\tau_{\mathrm{mid}} = p_0 + (\rho/2)(1-p_0)$. By (16), the post-attack mean drops below $\tau_{\mathrm{mid}}$ iff

$$\rho(1-\delta)^2(1-p_0) \;<\; \tfrac{\rho}{2}(1-p_0) \iff (1-\delta)^2 \;<\; \tfrac{1}{2},$$

which is independent of $(k, S, \rho, n)$ once $p_0 < 1$ (i.e., $k < S$). Solving gives $\delta^\star = 1 - 1/\sqrt{2}$. □

### A.5. Security against an oracle-blind adversary

**Theorem A.7** (Pseudorandomness under random oracle, oracle-blind adversary)**.** *Assume the secret key $\kappa$ has min-entropy at least $\lambda$ and $\mathcal{H}$ is modelled as a random oracle. Consider a polynomial-time adversary $\mathcal{A}$* without *access to the random oracle,* without *access to the LM (or its per-position distribution), and* without *access to a reference text drawn from the same prompt distribution. $\mathcal{A}$ has only the watermarked output $\mathbf{t}^w$ and the public scheme parameters $(S, T)$. Then $\mathcal{A}$ achieves:*

- *(i) key-recovery advantage at most $q \cdot 2^{-\lambda}$ for a query budget $q$ to any auxiliary oracle that depends on $\kappa$;*
- *(ii) state-prediction success at most $1/S + \mathrm{negl}(\lambda)$ on each fresh token $t^\star$ that has not been queried with the correct key, hence advantage $\mathrm{negl}(\lambda)$ over the uniform $1/S$ baseline;*
- *(iii) statistical distinguishing advantage at most $\mathrm{negl}(\lambda)$ between $\mathbf{t}^w$ and an i.i.d. uniform null $\mathbf{t}^r$.*

*Proof.* Without $\kappa$, the adversary's view of $\mathcal{H}(\kappa \,\|\, t)$ for any $t$ is uniform on the oracle output space. Recovery probability $q \cdot 2^{-\lambda}$ follows from a standard guessing argument over a min-entropy-$\lambda$ key. State prediction with advantage greater than $\mathrm{negl}(\lambda)$ would imply distinguishing the random-oracle output from uniform, contradicting (i).

For (iii): any statistic $\mathcal{D}(\mathbf{t})$ measurable in the token IDs alone that achieves non-trivial advantage $\varepsilon$ under the i.i.d. uniform null can be re-expressed as a function of $\sigma_\kappa$-dependent structure (since for an adversary with no LM access, the conditional distribution of $\mathbf{t}^w$ given $\sigma_\kappa$ is parameter-free in everything except the gate sequence and the partition). Such a function yields a state-predictor with advantage at least $\varepsilon$ on some fresh token, contradicting (ii) by a hybrid argument. Hence $\varepsilon \le \mathrm{negl}(\lambda)$. □

### A.6. Optimal LM-aware detector

**Theorem A.8** (Entropy-weighted detector is asymptotically Neyman–Pearson optimal)**.** *Suppose the verifier has access to the LM at each position, and let $g_i \in \{0,1\}$ denote the gate indicator at position $i$. Set $\pi_i = g_i + (1-g_i)/S$, the marginal probability of $X_i = 1$ under the watermark alternative. Treating $\{X_i\}$ as conditionally independent given the keyed state sequence — asymptotically valid under the random oracle on distinct tokens — the log-likelihood ratio*

$$\Lambda \;=\; \sum_{i=1}^{n-1}\Big[X_i \log(\pi_i S) + (1-X_i)\log \tfrac{1-\pi_i}{1-1/S}\Big] \tag{17}$$

*is asymptotically Neyman–Pearson optimal under the product-form approximation among all size-$\alpha$ tests with the known entropy profile.*

*Proof.* Under the null, $X_i \sim \mathrm{Bernoulli}(1/S)$ marginally with pairwise zero covariance (Theorem A.1). Under the watermark alternative, $X_i \sim \mathrm{Bernoulli}(\pi_i)$. The sequence $\{X_i\}$ is 1-dependent: $X_i$ and $X_{i+1}$ share $t_{i+1}$, but for $|i-j| \ge 2$, $X_i \perp X_j$ under the random-oracle model on distinct tokens.

The product-form likelihood ratio is

$$\begin{aligned}\Lambda &= \sum_i \log \frac{\pi_i^{X_i}(1-\pi_i)^{1-X_i}}{(1/S)^{X_i}(1-1/S)^{1-X_i}} \\ &= \sum_i X_i \log(\pi_i S) + (1-X_i)\log \tfrac{1-\pi_i}{1-1/S},\end{aligned}$$

matching the theorem statement (with the convention $0 \log 0 = 0$ for terms with $\pi_i = 1$, i.e., gated positions where $X_i = 1$ is deterministic). The Neyman–Pearson lemma applied to the product hypothesis identifies $\Lambda$ as uniformly most powerful for the product law.

By the central limit theorem for 1-dependent sequences (and vanishing pairwise covariance from Theorem A.1), the product-form $\Lambda$ has the same Gaussian limit as the true joint LLR up to vanishing $O(1/n)$ corrections in the local asymptotic normal (LAN) sense (Le Cam). Hence the test based on $\Lambda$ achieves the Neyman–Pearson power asymptotically. □

### A.7. Detector pseudocode

We restate the model-free detection algorithm for self-contained reading; the body version is 2.

The runtime is $O(n)$ hash evaluations and a single pass over the token stream; no LM access is required. The transition $T$ is the same matrix used at generation time, so the random baseline $p_0$ is computed directly from $T$ (clockwork: $p_0 = 1/S$; soft-cycle: $p_0 = 2/S$; general $k$-regular: $p_0 = k/S$).

**Algorithm 3** MCL Watermark Detection (model-free)

**Input:** text $x$, key $\kappa$, states $S$, transition $T$, threshold $\tau$
$\mathbf{t} \leftarrow \text{Tokenize}(x)$; $c \leftarrow 0$; $n \leftarrow |\mathbf{t}|$
**for** $i = 1$ **to** $n - 1$ **do**
  $s_i \leftarrow \mathcal{H}(\kappa \,\|\, t_i) \bmod S$; $s_{i+1} \leftarrow \mathcal{H}(\kappa \,\|\, t_{i+1}) \bmod S$
  **if** $T[s_i, s_{i+1}] > 0$ **then**
    $c \leftarrow c + 1$
  **end if**
**end for**
$\phi \leftarrow c/(n-1)$; $p_0 \leftarrow (\sum_{s,s'} T[s, s'])/S^2$
$z \leftarrow (\phi - p_0)/\sqrt{p_0(1-p_0)/(n-1)}$
**return** $(\mathbf{1}[\phi > \tau],\ \phi,\ z,\ 1 - \Phi(z))$

### A.8. Supplementary results: quality cost and self-healing

These two results were stated in earlier drafts; we retain the formal statements and proofs in the appendix for completeness, since they are referenced from elsewhere in the paper.

**Theorem A.9** (Quality cost identity and Jensen lower bound)**.** *At a gated position with current state $s$ and partition mass $Z_s = p(\mathcal{V}_{(s+1) \bmod S}) > 0$, the KL between the renormalised MCL distribution $P_{\text{MCL}}(t) = p(t)/Z_s \cdot \mathbf{1}[t \in \mathcal{V}_{(s+1) \bmod S}]$ and the LM distribution $p$ is $D_{\text{KL}}(P_{\text{MCL}} \,\|\, p) = \log(1/Z_s)$. Under hash uniformity, $\mathbb{E}[Z_s] = 1/S$, and Jensen's inequality gives*

$$\mathbb{E}[D_{\text{KL}}] \ \geq\ \log S, \tag{18}$$

*with equality iff $Z_s$ is constant in the key. Averaged over positions with gate rate $\rho$, $\mathbb{E}[D_{\text{KL}}]_{\text{tok}} \geq \rho \log S$.*

*Proof.* Direct computation: $D_{\text{KL}}(P_{\text{MCL}} \,\|\, p) = \sum_{t \in \mathcal{V}_{(s+1) \bmod S}} (p(t)/Z_s) \log(1/Z_s) = \log(1/Z_s)$. Under hash uniformity, $\Pr[t \in \mathcal{V}_{(s+1) \bmod S}] = 1/S$ for every fixed $t$, so $\mathbb{E}[Z_s] = \sum_t p(t)/S = 1/S$. Concavity of log and Jensen give $\mathbb{E}[\log Z_s] \leq \log \mathbb{E}[Z_s] = -\log S$, hence $\mathbb{E}[\log(1/Z_s)] \geq \log S$. The token-averaged bound follows by averaging over positions with gate rate $\rho$. □

**Theorem A.10** (Self-healing against oracle-blind adversaries)**.** *Let $\mathcal{T}_{\text{q}} \subseteq \mathcal{V}$ be the set of tokens the adversary has queried with the correct key. For any token $t' \notin \mathcal{T}_{\text{q}}$ that the adversary introduces, $\sigma_\kappa(t')$ is uniform on $[S]$ by the random-oracle property, so each modified position contributes an indicator distributed as* $\text{Bernoulli}(1/S)$ *to the validity count, independently of strategy. Hence the expected mass contributed to $\phi$ by modified-pair positions satisfies*

$$\Sigma(\delta) \ \geq\ \tfrac{1}{S}\, \delta(2 - \delta), \tag{19}$$

*with equality when every introduced token is unqueried.*

*Proof.* The fraction of token pairs $(t_i, t_{i+1})$ touching at least one modified token is $1 - (1-\delta)^2 = \delta(2-\delta)$. Each such pair contributes an indicator with conditional expectation $1/S$ by the same column- and row-regularity argument as in Theorem A.3. Strategies in which the adversary introduces previously-queried tokens (whose state is in $\sigma_\kappa(\mathcal{T}_{\text{q}})$) can craft pairs that hit valid transitions deterministically, raising $\phi$ above $\Sigma(\delta)$ — which only helps detection. The lower bound $\Sigma(\delta) \geq \delta(2-\delta)/S$ is therefore strategy-free. □

## B. Reference Tables

### B.1. Closed-Form Calibration Lookup

*Table 5.* Calibration lookup $S^\star(n, \rho, \alpha)$ from Theorem A.2. Rows are text lengths; columns are $(\alpha, \rho)$ pairs. Entries are derived analytically from the closed-form detection bound and do not require empirical anchoring.

| | $\alpha = 10^{-3}$ | | $\alpha = 10^{-6}$ | |
|---|---|---|---|---|
| $n$ | $\rho$=0.3 | $\rho$=0.5 | $\rho$=0.3 | $\rho$=0.5 |
| 100 | 6 | 3 | 12 | 5 |
| 200 | 4 | 2 | 7 | 3 |
| 500 | 2 | 2 | 4 | 2 |
| 1000 | 2 | 2 | 3 | 2 |

## C. Experimental Protocol and Reproducibility

This appendix records every setting needed to reproduce the head-to-head matched-budget study of section 6, the per-cell tables in the main results, and every figure.

### C.1. Models, tokenizers, and hardware

**Model fleet.** The headline experiments use three instruction-tuned 7–8B parameter open-weight LLMs, each loaded via the Hugging Face `transformers` library. The exact `repo_id`, revision pin, and `torch_dtype` for every model are listed in the public code release alongside its prompt template; we do not enumerate them here in order to keep the discussion model-agnostic. Each model is run on a single NVIDIA GPU (H100 or H200, depending on memory pressure); CPU fallback is supported for the detector but not for generation. Every model uses its own native tokenizer, both for generation and for detection, and self-perplexity (PPL) is computed on the same model's logits that produced the text.

**Decoding.** Watermarked positions use greedy argmax over the allowed-partition mask of 1; ungated positions use temperature sampling at $T = 0.7$. The same temperature is used to draw pilot generations for gate-threshold calibration.

**Gated models.** A subset of the fleet is gated on the Hugging Face Hub. The released environment expects an HF_TOKEN in scope at runtime; the token is read by huggingface_hub and never logged to disk. We do not name specific gated repositories in this appendix; the public code drop records each repo_id alongside its license terms.

### C.2. Domains and prompts

**Four domains, $N = 200$ prompts each.** Each domain runs on $n = 200$ prompts drawn from a fixed pool, identical across gates, models, and attacks within a cell.

- **Code:** HumanEval Python signatures with their natural-language docstrings as the prompt; the generation is the function body.
- **Factual:** short closed-answer prompts asking for a single attested fact (capitals, dates, named entities).
- **Wiki:** open-ended descriptive prompts derived from a curated Wikipedia concept list, expanded via a fixed template.
- **Writing:** open-ended creative-writing prompts requesting a short narrative or argumentative passage.

Every prompt set, with template strings, prompt indices, and a deterministic shuffle seed, is shipped under data/v7_min/<domain>/records.jsonl.

### C.3. MCL hyperparameters

**Default cell.** $S = 5$ states, clockwork transition $T(s, s') = \mathbb{1}[s' \equiv s{+}1 \pmod S]$, watermark budget $\rho = 0.5$, secret key fixed across all cells, runs, and models.

**State sweep.** subsection 6.5 sweeps $S \in \{2, 3, 5, 7, 11\}$ at fixed $\rho = 0.5$ to anchor the closed-form calibration of Theorem A.2 on data.

**$k$-regular topology.** subsection 6.4 runs the soft-cycle ($k = 2$) topology at the default cell and compares the empirical robustness threshold against $\delta^\star$ to validate Theorem A.5.

### C.4. Baseline calibration

The two baselines, KGW (Kirchenbauer et al., 2023) and SWEET (Lee et al., 2024), are matched to MCL's budget at the cell level. SWEET is an entropy gate that activates the watermark only at *high*-entropy positions (the structural opposite of schemes that gate at low-entropy positions); we configure it so that the realised gate rate over the pilot pool is within $\pm 0.02$ of $\rho = 0.5$, by quantile-matching $\tau_H$ to the $(1 - \rho)$ quantile of the per-position entropy distribution. KGW runs at the same effective budget by construction, since it gates every position. All three schemes share the same secret key and the same generation pool.

### C.5. Attack protocol

Each watermarked generation is subjected to three independent attack streams; post-attack detection statistics are reported per cell in section 6.

- **Random substitution** ($\delta = 0.20$): $\lceil 0.20\, n \rceil$ token positions chosen uniformly at random and replaced with uniform draws from the model's tokenizer vocabulary.
- **Translation round-trip:** EN $\to$ $L$ $\to$ EN via the NLLB-200 distilled model (NLLB Team, 2022), where $L \in \{\text{French}, \text{German}, \text{Russian}, \text{Chinese}\}$. Each language constitutes a separate cell.

We do not include grammar-preserving paraphrase attacks in this release; their evaluation is deferred to the extended version.

### C.6. Randomisation

random, numpy.random, and torch.manual_seed are all seeded to 42 at the start of each generation run. The random-substitution attack uses an independent seed (43). Bootstrap resamples in section 6 use a per-contrast key derived deterministically from the cell descriptor.

### C.7. Released artefacts

For each generation we record the prompt, the watermarked output, the per-position gate signal, the realised gate rate $\bar{\rho}$, PPL, the detector statistic $\phi$ and its $z$-score, and every post-attack $\phi$, $z$, and detection flag. All tables and figures in this paper derive from these records. The code drop ships under the same repository as this paper; the per-domain JSONL records under data/v7_min/<domain>/records.jsonl are the source-of-truth for every empirical claim.

## D. Additional Run Data and Open Extensions

### D.1. Translation per-pivot breakdown

### D.2. FPR recalibration: SD recipe and failure modes

### D.3. Apples-to-apples head-to-head at empirical FPR$= 1\%$

### D.4. Empirical FPR$= 0\%$ achievability

**Open data and deferred extensions.** The following experiments are deferred to the extended version of this paper:

- **DIPPER paraphrase attack.** Grammar-preserving paraphrase via DIPPER (Krishna et al., 2023) on at

*Table 6.* **Per-pivot MCL translation robustness.** TPR @ $z > 2.326$ for each NLLB-200 pivot under MCL $S = 5$, $\rho = 0.5$. Llama and Qwen cover four domains (factual, wiki, writing, code) at 25 prompts each ($n = 100$ per row); the Mistral row covers only three domains (factual, wiki, writing; $n = 75$). ZH is the most aggressive pivot ($\delta_{\text{eff}} \approx 0.81$); FR, DE, RU are gentler ($\delta_{\text{eff}} \approx 0.69$).

| Model | $n$ | clean | FR | DE | RU | ZH |
|---|---|---|---|---|---|---|
| Llama-3.1-8B | 100 | 100% | 85% | 78% | 81% | 75% |
| Mistral-7B | 75 | 100% | 88% | 92% | 92% | 75% |
| Qwen-2.5-7B | 100 | 100% | 78% | 75% | 78% | 66% |

*Table 7.* **Empirical-SD recalibration vs. failure modes, evaluated on $n = 3000$ non-watermarked samples (1000 per model).** Goal: bring empirical FPR close to the $\alpha = 1\%$ target while preserving TPR$= 100\%$. Only the empirical-SD recipe (Fix 1) brings FPR within $\sim 0.2$ pp of the target ($1.17\%$); the other four sit at 2–5% FPR or collapse TPR. The iid-baseline row is the closed-form $z_\alpha = 2.326$ threshold without recalibration. The $2.07\%$ baseline on this $n = 3000$ pooled corpus supersedes the 1.7–6.9% per-$S$ cells in Table 4 (which use the smaller Exp. 5 sub-design with $n \approx 300$ per cell).

| Method | FPR | TPR |
|---|---|---|
| iid baseline (no fix; $z > 2.326$) | 2.07% | 100.0% |
| **Fix 1: empirical SD recal.** | **1.17%** | **100.0%** |
| *Failure modes (reported, not adopted):* | | |
| Fix 2: Newey–West HAC | 3.45% | 100.0% |
| Fix 3: $k$-skip ($k = 3$) | 2.46% | 2.2% |
| Fix 4: stopword filter (top-200) | 5.18% | 99.0% |
| Fix 5: HDD-lite (inv.-freq. weighted) | 2.35% | 100.0% |

least one (model, domain) cell.

- **$k$-regular validation at $k \geq 3$.** Theorem A.5 predicts $p_0 = k/S$ for any $k$-regular topology; we validate $k \in \{1, 2\}$ only.
- **External-LM PPL or MAUVE quality metric.** Self-PPL is conservative but circular; an external judge LM or MAUVE would give a model-independent quality reading.
- **Base (non-instruction-tuned) and $\geq 10$B LLMs.** Our fleet covers only instruction-tuned 7–8B models.
- **$\rho$-sweep on the new fleet.** The current headline runs use $\rho = 0.5$ only; the previous gate-invariance ablation was on a smaller setup.
- **Adversary with detector-oracle access.** Theorem A.7 excludes adversaries who can query the detector; characterising query complexity is open.

## E. Reference Detector Implementation

The following 24-line Python listing is a self-contained, dependency-minimal reference implementation of 2 for

*Table 8.* **Empirically-calibrated head-to-head at FPR$= 1\%$.** Each method's threshold is the maximum across the per-model 99% $z$-quantiles on a non-watermarked calibration corpus (the conservative recipe; per-model nulls in MCL: Llama 2.87, Qwen 2.87, Mistral 2.41, so MCL $z^\star = 2.87$; KGW $z^\star = 2.62$). MCL retains its TPR advantage at matched empirical FPR, confirming the analytical-threshold comparison in Table 1 is not an FPR artefact. SWEET is omitted from this table: the SWEET null $z$-distribution was not collected on the FPR corpus and cannot be reproduced from the released artefacts; an SWEET-recalibrated row is deferred to the extended version.

| Method | $\text{TPR}_{\text{clean}}$ | $\text{TPR}_{\text{random}}$ | $\text{TPR}_{\text{ZH}}$ |
|---|---|---|---|
| **MCL $S = 5$ (ours)** | **100.0%** | **99.9%** | **68.7%** |
| KGW $\gamma = 0.5$ | 69.0% | 45.4% | 15.3% |

*Table 9.* **FPR$= 0\%$ by lifting the threshold to $z^\star = \max_i z_{\text{nwm},i} + 0.5$.** Computed on the held-out non-watermarked corpus ($\sim 1000$ samples per model). Watermarked text retains TPR$= 100\%$ because clean MCL $z$ at the canonical $S = 5$ ($\approx 25$ across the three models; Table 4) is several times the maximum observed null $z$. KGW and SWEET cannot achieve this regime because their watermarked and non-watermarked $z$-distributions overlap.

| Model | max $z_{\text{nwm}}$ | lifted $z^\star$ | TPR @ $z^\star$ |
|---|---|---|---|
| Llama-3.1-8B | 7.83 | 8.33 | 100% |
| Qwen-2.5-7B | 5.00 | 5.50 | 100% |
| Mistral-7B | 4.82 | 5.32 | 100% |

clockwork MCL. It takes a list of integer token IDs, a bytes key, a state count $S$, and a significance threshold $\alpha$; it returns the fingerprint score $\phi$, the $z$-score, the one-sided $p$-value, and a binary detection flag.

The listing is sufficient to independently verify any MCL-watermarked text given only the secret key, with no language-model access and no learned components. Extensions to arbitrary $k$-regular topologies (cf. Theorem A.5) require replacing the successor check `states[i+1] == (states[i]+1) % S` with a lookup `T[states[i]][states[i+1]] > 0` and setting `p0 = (T > 0).mean()`; no other modification is needed.

**Complexity.** Two SHA-256 evaluations per token ($2n{\cdot}256$ bits of hash output) and $n - 1$ integer equality checks. At $n = 100$ the entire detection pipeline runs in under 1 ms on a single CPU core; no GPU, no model weights, no tokenizer aside from what is needed to obtain `token_ids`.

**Interpretation as an audit primitive.** The detector's inputs are exactly what a third-party auditor could receive under an Article 50 disclosure regime: a public text, a detector program, and a (confidentially held) key. The output is a $p$-value with analytically known false-positive behaviour under the null, sidestepping the calibration-by-grid-search

```
import hashlib
from math import sqrt
from scipy.stats import norm

def sha_state(key: bytes, token_id: int, S: int) -> int:
    """SHA-256-based state assignment sigma_kappa(t) = H(kappa || t) mod S."""
    h = hashlib.sha256(key + token_id.to_bytes(8, "big")).digest()
    return int.from_bytes(h[:8], "big") % S

def detect_mcl(token_ids, key: bytes, S: int,
               alpha: float = 0.01) -> dict:
    """Clockwork MCL detector; O(n) in the number of tokens."""
    n = len(token_ids)
    if n < 2:
        return {"phi": 0.0, "z": 0.0, "p_value": 1.0,
                "is_watermarked": False}
    states = [sha_state(key, t, S) for t in token_ids]
    valid = sum(1 for i in range(n - 1)
                if states[i + 1] == (states[i] + 1) % S)
    phi = valid / (n - 1)
    p0 = 1.0 / S
    se = sqrt(p0 * (1.0 - p0) / (n - 1))
    z = (phi - p0) / se
    p_value = 1.0 - norm.cdf(z)
    return {"phi": phi, "z": z, "p_value": p_value,
            "is_watermarked": p_value < alpha}
```

*Figure 6.* Reference detector for clockwork MCL: 24 lines of dependency-minimal Python (`hashlib` + `scipy.stats`). Inputs are the tokenizer's token IDs, the secret key, and the state count; outputs are the fingerprint $\phi$, $z$-score, one-sided $p$-value, and a binary detection flag.

problem that afflicts logit-bias watermark families.

**Key management.** The scheme's security reduces to the confidentiality of `key` plus the random-oracle idealisation of SHA-256 (Theorem A.7). In practice an auditor and a deployer can share the key through any standard key-management substrate (e.g., HKDF-derived per-deployment keys, committed to an external ledger so that the commitment precedes generation). Rotating keys per deployment epoch preserves the $p_0 = k/S$ null baseline without changing any detector behaviour.